BRAC UNIVERSITY **Set: A**

Department of Computer Science and Engineering

Examination: Mid Term	Semester: Fall 2025
Duration: 1 Hour 25 Minutes	Full Marks: 25

CSE 260: Digital Logic Design

Answer the following questions. Show calculations where required.

Figures in the right margin indicate marks. Understanding the question is part of the examination.

1.CO2 The **"Top Tier"** team is building a system for a spacecraft that aims to detect two specific signals: **Signal Alpha** and **Signal Omega**. The spacecraft is equipped with four types of pulse detection sensors: **P, Q, R, and S**. The system processes the pulse values from these sensors and detects **Signal Alpha** and **Signal Omega** based on the pulse patterns.

Inputs:

The system has four pulse detection sensors (P, Q, R, S). For each sensor:

- Input = 1 if a pulse is detected.
- Input = 0 if no pulse is detected.

Outputs:

- Signal Alpha (A) = 1 if detected.
- Signal Omega (O) = 1 if detected.

Conditions for Detecting Signal Alpha:

i) If the decimal value of the pulse sequence of the four sensors (**P, Q, R, S**) results in an **even number** in the **Fibonacci series**, then Signal Alpha is detected. Check below if you don't understand the **Fibonacci series**.

- The **Fibonacci series** starts as: 0, 1, 1, 2, 3, ... (i.e., each number is the sum of the previous two numbers).
- The **even Fibonacci numbers** in the series are: 0, 2, 8..

ii) If the decimal value of the pulse sequence of the four sensors results in an **odd number** in the Fibonacci series, then the spacecraft's detection module cannot correctly identify Signal Alpha. In this case, the output for Signal Alpha is **Don't Care**.

iii) For all other combinations of pulses, Signal Alpha is **not detected**.

Conditions for Detecting Signal Omega:

i) If there are **consecutive 1s** in the pulse sequence (i.e., two or more 1s occurring back-to-back), and the resulting number is not odd (i.e., 1100 equals 12 which is even), then Signal Omega is detected.

ii) If there are **consecutive 0s** (i.e., two or more 0s occurring back-to-back) in the pulse sequence, and Pulse P is detected, then Signal Omega is detected.

iii) Signal Omega **cannot be detected** if Signal Alpha has already been detected for the same combination of pulses. In this case, the output for Signal Omega is **Don't Care** (whether the conditions for Signal Omega are met or not).

iv) For all other combinations of pulses that do not meet the above criteria, Signal Omega is **not detected**.

a) **Create** a truth table for the system described above. For the outputs, use the variables **A** and **O**, and use **Don't Care** values where necessary. For the inputs, maintain the order as **P**, **Q**, **R**, S, where **P** is the MSB and **S** is the LSB. 6

b) **Use** Karnaugh Map(s) to find out the simplified expressions for each of the outputs. 4

2. CO1 a) You are designing a temperature monitoring system for a spacecraft. You received two sample temperatures,

Sensor A: $-(235)_{10}$ [this is negative two hundred and thirty five in decimal]

Sensor B: $-(252)_{10}$ [this is negative two hundred and fifty two in decimal]

i) Convert the two integers into 10 bit 2's complement numbers. Then add and compute the result using the 10-bit 2's complement number system. State and justify whether overflow occurs or not. Show all conversions. 3+1

Now, the total heat load for the spacecraft results in $(131415)_6$ which is later distributed equally among $(29)_{10}$ cooling units.

ii) Determine how many units of load each cooling unit receives for the total heat load. You must use only base 6 for division. 2

3. CO1 a) **Simplify** the following function to the minimum number of literals using boolean algebraic formulas and draw the circuit diagram of the simplified equation: 4+1

$$F(P, Q, R, S) = ((PQ)\oplus(RS))(S'+P'QS)S'(Q\odot R) + ((P+S'+Q)(S'+(Q+S)')R)'$$

NB: You cannot use a truth table or K-map to solve this question.

b) **Convert** the following function to canonical POS using distributive law and mention the maxterms: 4

$$F(P, Q, R, S) = P'Q+PR+QS'$$

NB: You cannot use truth table or K-map to solve this question, you also cannot do canonical SOP first and then convert to canonical POS later.